

US mathematician, computer scientist, and naval officer Grace Hopper was a pioneer in the field of computer technology. She helped to develop the earliest computer languages, which enabled computers to be programmed with ordinary words.

MILESTONES

HARVARD POSTING

Accepts naval posting to Harvard in 1944; develops and programs the Harvard Mark I computer.

DEVELOPS UNIVAC I

Joins Eckert-Mauchly Corp. in 1949 to help devise first commercial electronic computer.

COMPILER PROGRAM

Develops Flow-Matic, the first English-language data-processing compiler, with her team in 1955.

COMPUTER LANGUAGE

In 1959, co-develops COBOL, the first standard computer language, and promotes its global adoption.

GLOBAL RECOGNITION

Receives many awards and honors, including the first Man of the Year in Computer Science in 1969.

After graduating from Vassar College, New York, Grace Hopper attended Yale University and gained her PhD in mathematics in 1934. She joined the US Naval Reserves in 1943. During her long career, Hopper would attain the position of rear admiral in the navy and become one of the foremost computer programmers of her generation.

Pioneering technology

In 1944, Hopper was posted to Harvard University, where she helped develop the Mark series of computers. This began with Mark I, which was one of the first electromechanical computers and a prototype of the electronic computer. In 1949, she joined the Eckert-Mauchly Computer Corporation to help build UNIVAC 1, the earliest computer designed for business and administrative use in the US. Soon after, she improved the computer compiler—a program that translated symbolic mathematical instructions into codes that a computer could read directly. This was a key step in the development of computer languages.

Hopper championed the idea of making computers accessible to a nontechnical market, and in 1955, she designed Flow-Matic, the first programming language that responded to English word commands. She then helped develop, refine, and promote COBOL, a business-orientated language that became widely used. With her visionary skills and outstanding technical ability, Hopper simplified programming and enabled broad commercial usage of computer technology.

Unlike her contemporaries, Grace Hopper understood the potential commercial applications of computers and made such usage possible through her ongoing work on compilers.

